

COASTAL LANDFORMS / INDIA COAST AND CRZ

COASTAL-SYSTEM BALANCE → REFRACTION MAP → EROSIONAL SEQUENCE → CLIFF RETREAT SECTION → DEPOSITIONAL DISCRIMINATOR → RELATIVE SEA-LEVEL FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 COASTAL-SYSTEM BALANCE

COASTAL-SYSTEM BALANCE WAVES + TIDES + CURRENTS MARINE ENERGY RIVERS + CLIFFS + OFFSHORE SEDIMENT SUPPLY SEA LEVEL + LAND MOTION ACCOMMODATION SPACE HUMAN WORKS

WAVES + TIDES + CURRENTS → marine energy

RIVERS + CLIFFS + OFFSHORE → sediment supply

SEA LEVEL + LAND MOTION → accommodation space

HUMAN WORKS → redistribute erosion and accretion

MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and

CORE CLAIM

- WAVES + TIDES + CURRENTS → marine energy
- RIVERS + CLIFFS + OFFSHORE → sediment supply

MECHANISM

- SEA LEVEL + LAND MOTION → accommodation space
- HUMAN WORKS → redistribute erosion and accretion

CONSEQUENCE / LIMIT

- MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and
- sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

01

STAGE 01 REFRACTION MAP

REFRACTION MAP DEEP WATER WAVE FRONTS TRAVEL FASTER SHALLOW HEADLAND FRONTS SLOW AND CONVERGE FRONTS DIVERGE AND ENERGY DISPERSES EROSION FOCUS DEPOSITIONAL SHELTER

DEEP WATER → wave fronts travel faster

SHALLOW HEADLAND → fronts slow and converge

BAY → fronts diverge and energy disperses

RESULT → erosion focus / depositional shelter

CORE CLAIM

- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

MECHANISM

- BAY → fronts diverge and energy disperses
- RESULT → erosion focus / depositional shelter

CONSEQUENCE / LIMIT

- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY → fronts diverge and energy disperses; RESULT → erosion focus / depositional shelter.

02

STAGE 02 EROSIONAL SEQUENCE

EROSIONAL SEQUENCE HYDRAULIC ACTION + ABRASION WIDENING AND ROOF ATTACK ROOF COLLAPSE

JOINT / FAULT → hydraulic action + abrasion

CAVE → widening and roof attack

ARCH → roof collapse → STACK

STACK → undercutting → STUMP

CORE CLAIM

- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

MECHANISM

- ARCH → roof collapse → STACK
- STACK → undercutting → STUMP

CONSEQUENCE / LIMIT

- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.

03

STAGE 03 CLIFF RETREAT SECTION

CLIFF RETREAT SECTION WAVE NOTCH BASAL UNDERCUTTING INSTABILITY AND COLLAPSE DEBRIS REMOVAL RENEWED ATTACK WIDENING WAVE-CUT PLATFORM

CORE CLAIM

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

MECHANISM

- DEBRIS REMOVAL → renewed attack
- RETREAT → widening wave-cut platform

CONSEQUENCE / LIMIT

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.



